OFFICIAL SYSTEM SPECIFICATION • ARCHITECTURE REPORT

SPEC-2025-11-25

TableTalk

High-Level Design, Low-Level Concurrency Architecture, Sequence Flows, User Journeys & Persona Specifications for a Production Model Context Protocol (MCP) Server.

Rust 1.80+ | Axum 0.8 | Tokio Async | MCP Streamable HTTP | SQLite 3 WAL | BEGIN IMMEDIATE | Groq Llama 3.3 70B | AWS Bedrock Nova Lite | React 19 / Vite

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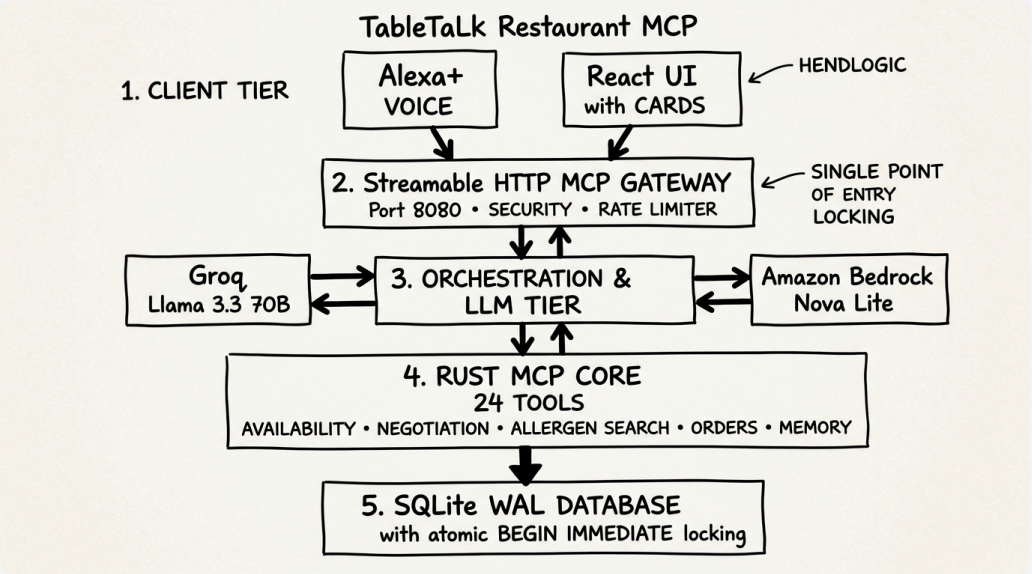
PROTOCOL
Model Context Protocol

TARGET TRACK
Amazon Developer (Alexa+)

TOOLING BREADTH
24 Validated MCP Tools

High-Level System Design (HLD)

5 Decoupled Architectural Tiers with Defense-in-Depth Security



1. Client Tier (Alexa+ & React UI)

Dual-output conversational frontend. Emits voice SSML for smart speakers while generating accessible cards from server `structuredContent`.

2. Streamable HTTP MCP Gateway (:8080)

Single multiplexed `/mcp` endpoint replacing fragile dual-port architectures. Enforces origin checks, token-bucket IP rate limits, and panic isolation.

3. Conversational AI & Orchestrator

Dynamic tool-calling client using Groq Llama 3.3 70B for <300ms time-to-first-token and Amazon Bedrock Nova Lite. Hermetic scripted fallback guarantees zero demo hallucination.

4. Rust MCP Core (24 Tools)

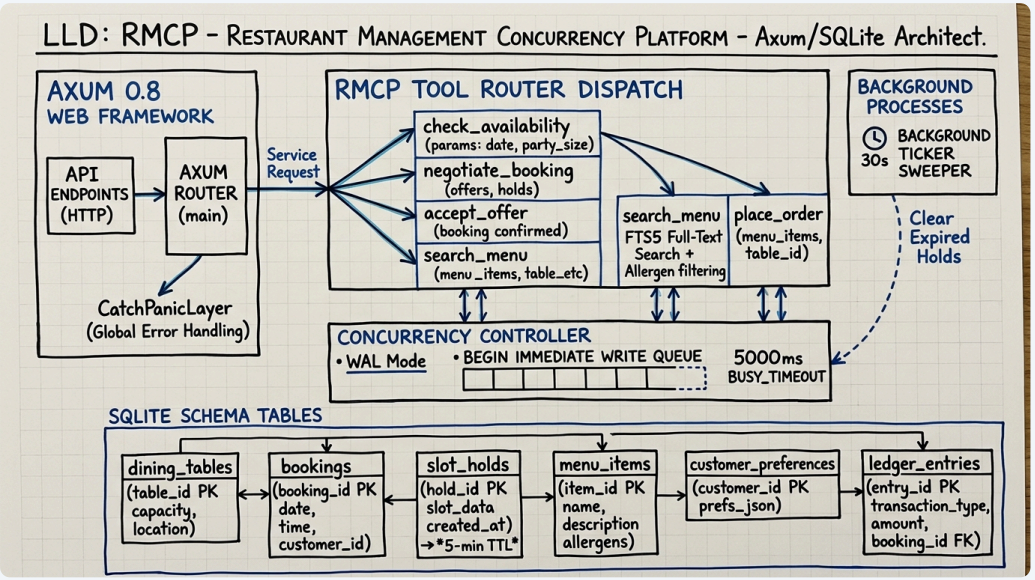
Strongly typed domain tool modules covering capacity checks, table negotiation, 5-minute atomic holds, allergen filtering, pickup orders, and privacy preferences.

5. SQLite WAL Storage Subsystem

Zero-overhead embedded persistence with Write-Ahead Logging and `BEGIN IMMEDIATE` locks preventing double-bookings during concurrent traffic spikes.

Low-Level Design & Concurrency Architecture

Axum Dispatcher, Concurrency Controller & Relational Model



Axum 0.8 & RMCP Dispatch Pipeline

Incoming JSON-RPC requests traverse `CatchPanicLayer`, `validate_origin`, and parse into strongly-typed Rust structs using `schemars` schemas.

BEGIN IMMEDIATE Locking

Standard SQLite `DEFERRED` locks risk runtime deadlocks. TableTalk enforces `BEGIN IMMEDIATE` with 5000ms busy timeout to serialize table hold writes.

30s Ticker Sweeper

Background Tokio cron task purges expired 5-minute table holds (`slot_holds`) and unconfirmed deposit sessions automatically.

FTS5 Allergen Index

Virtual SQLite FTS5 table links ingredients to 10 major allergen classes, enforcing strict cross-contact warnings.

Integer Minor Units

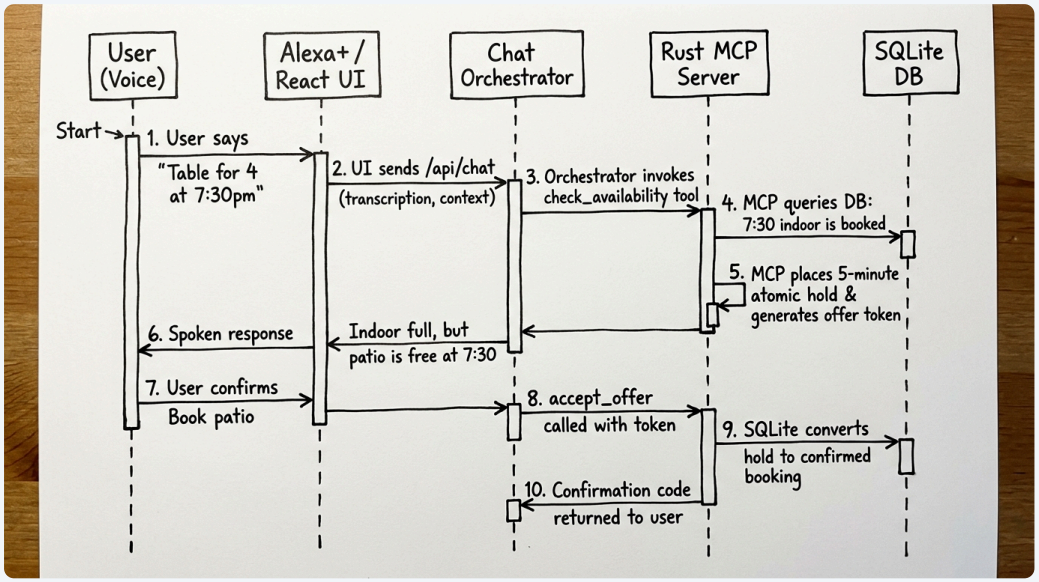
Currencies stored in integer paise/cents. Overflow-checked arithmetic eliminates floating-point drift.

Primary Relational Entities

Table	Key Role	Concurrency Control
`dining_tables`	Capacity & Zone registry (indoor, patio, bar)	Read-heavy, cached in memory
`slot_holds`	5-minute atomic holds with cryptographic token	Atomic write lock + 30s TTL sweep
`bookings`	Confirmed reservations with deposit state	Unique time & table constraint index
`customer_preferences`	Opt-in dining memory (dietary & allergies)	SHA-256 hashed token, hard erasure

Conversational Negotiation Sequence Flow

Hero Flow: Peak Hour Capacity Collision & Atomic Hold Placement



Step 1–3: Request Ingestion & Availability Check

User speaks *"Table for 4 at 7:30 PM indoor"*. Alexa/UI transmits request to `/api/chat`. Orchestrator executes `check_availability` tool.

Step 4–5: Collision Detection & Hold Generation

SQLite queries reveal all indoor 4-top tables (`T03`, `T04`, `T05`) are booked. Server ranks alternatives (Patio at 7:30 PM, Indoor at 9:00 PM), writes a 5-minute row to `slot_holds`, and issues an offer token.

Step 6–7: Explanatory Spoken Dialogue

Assistant speaks: *"I don't have indoor seating at 7:30 PM, but patio is free at 7:30 PM, or 9:00 PM indoor. Would either work for you?"* Diner responds: *"Book patio."*

Step 8–10: Atomic Conversion & Confirmation

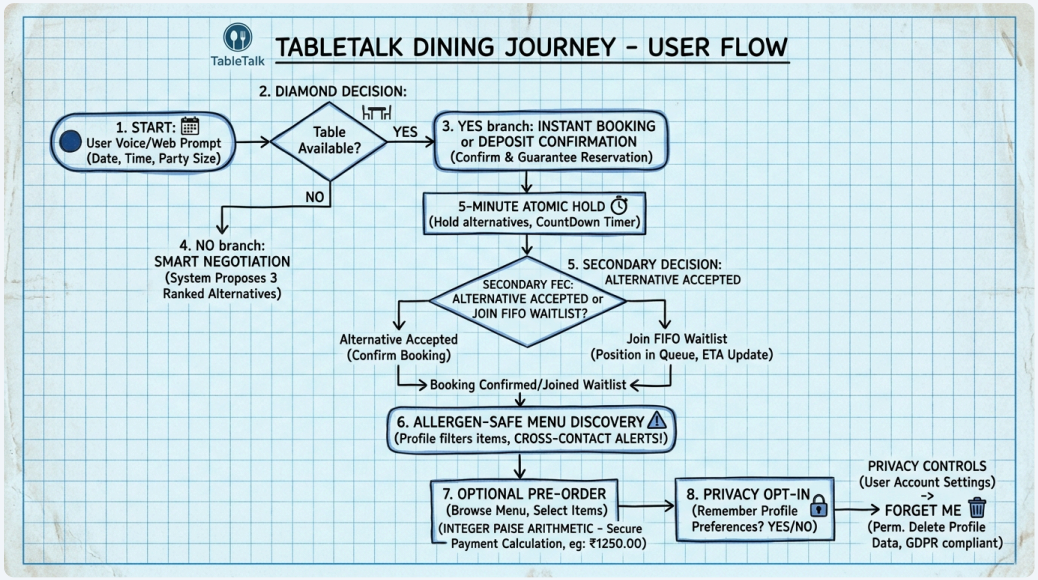
Orchestrator invokes `accept_offer` with the hold token. Database atomically deletes the hold and inserts the booking with code `TT-8102`. UI renders confirmed card.

Anti-Snatching Guarantee

The 5-minute atomic hold prevents competing diners from claiming the offered patio table during the conversational turn.

End-to-End User Dining Journey

Discovery, Negotiation, Allergen Invariants & Privacy Controls



1. Booking & Negotiation Branch

When the desired slot is open, immediate confirmation is granted. When occupied, the user is offered 3 ranked alternatives under a 5-minute atomic lock, or given the option to join a FIFO waitlist.

2. Uncompromising Allergen Safety Invariant

- **No Absolute Claims:** Words like *"guaranteed safe"* or *"allergy-free"* are strictly filtered and prohibited.
- **Cross-Contact Flagging:** Discloses shared fryers and grill surfaces transparently.
- **Explicit Confirmation:** Diners must verbally confirm allergen notices before order placement.

3. The Human-Confirms Payment Rule

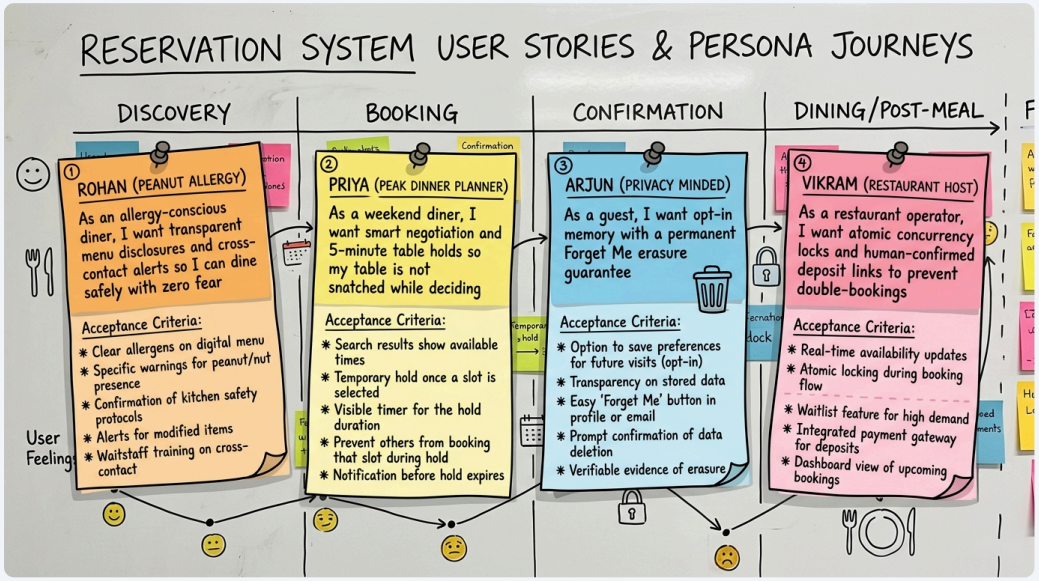
The AI assistant **never** collects card numbers or approves charges directly. For group deposits, it generates a secure external checkout link where the human confirms payment.

4. Privacy-Preserving Memory & Hard Erasure

- **Explicit Consent:** Preferences remembered only after verbal opt-in (`enable_memory`).
- **SHA-256 Tokens:** Identifiers stored strictly as salted hashes.
- **Forget Me Guarantee:** 1-click irreversible hard deletion of all customer preference records.

User Stories & Agile Acceptance Criteria

Persona Needs, Journey Touchpoints & Quality Gates



Rohan • Allergy-Conscious Diner

"As an allergy-conscious diner, I want transparent menu disclosures and cross-contact alerts so I can dine safely with zero fear."

- Allergen tags clearly highlighted on UI cards.
- Specific warnings for peanut/tree nut cross-contact.
- Confirmation required before finalizing order.

Priya • Peak Hour Dinner Planner

"As a weekend diner, I want smart negotiation and 5-minute table holds so my table is not snatched while deciding."

- Automated ranking of nearby alternative slots.
- 5-minute countdown hold on offered tables.
- No dead-end "fully booked" rejection responses.

Arjun • Privacy-Minded Guest

"As a guest, I want opt-in memory with a permanent Forget Me erasure guarantee."

- Opt-in consent prompt before storing dietary preferences.
- Immediate cryptographic purging on `forget_me` command.

Vikram • Restaurant Host & Manager

"As a restaurant operator, I want atomic concurrency locks and secure deposit links to prevent double-bookings."

- Zero double-bookings verified under load.
- Automated group deposit enforcement for parties of 6+.